NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

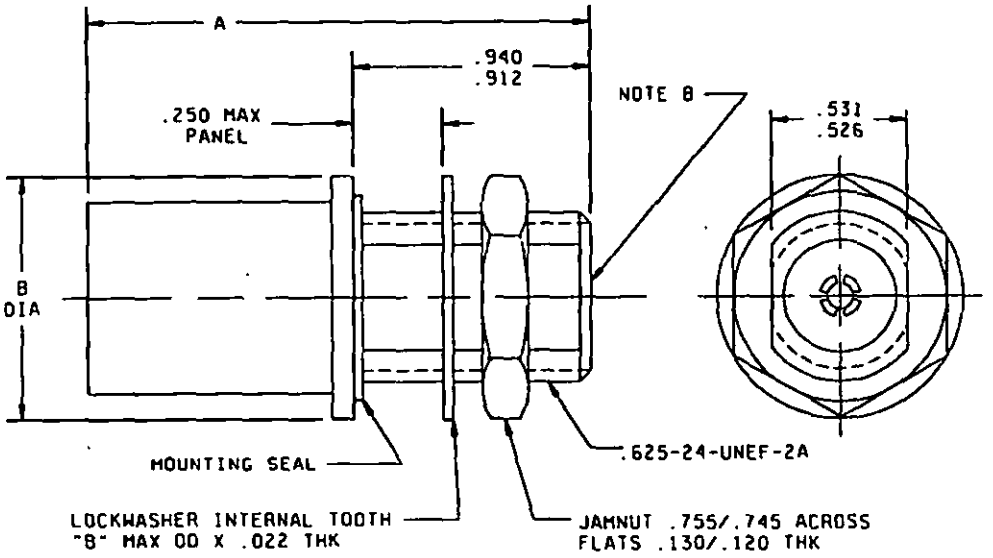
MIL-PRF-39012/3G
26 September 1994
SUPERSEDING
MIL-C-39012/3F
3 October 1986

PERFORMANCE SPECIFICATION SHEET

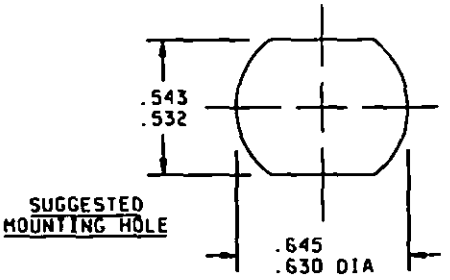
CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY
(SERIES N (CABLED), JAM NUT MOUNTED, SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-39012.



Inches	mm	Inches	mm
.022	0.56	.625	15.88
.120	3.05	.630	16.00
.130	3.30	.645	16.38
.250	6.35	.745	18.92
.526	13.36	.755	19.18
.531	13.49	.912	23.16
.532	13.51	.940	23.88
.543	13.79		

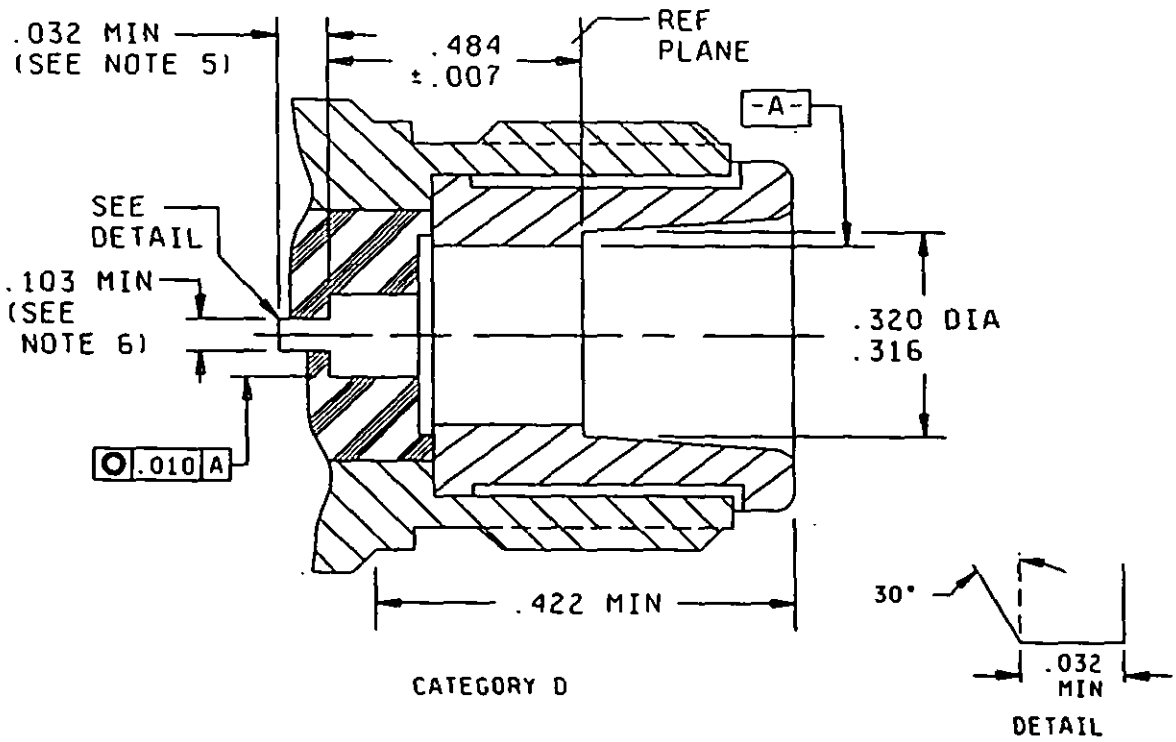


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. For dimensions A and B, see tables I and III.
4. Dimensions A and B are the largest overall dimensions of the connector.
5. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
6. All undimensioned pictorial representations are for reference purposes only.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
8. Series N, socket contact interface, in accordance with MIL-STD-348.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm
.007	0.18
.010	0.25
.032	0.81
.103	2.62
.316	8.03
.320	8.13
.422	10.72
.484	12.29

NOTES:

1. Dimensions are in inches.
2. Slot design is optional.
3. Full threads to within .063 (1.60 mm) of shoulder; 1 1/2 maximum uneven threads to shoulder.
4. Metric equivalents are given for information only.
5. Chamfer is optional, if chamfer is used put chamfer on a 30° maximum.
6. Dimension to meet connector performance requirements.

FIGURE 2. Category D captivation detail.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash number 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters)	
			Minimum	Maximum
Category A - Field serviceable (no special tools required) 3/				
0101 Cable group VIII	2-RG006 4/ 73-RG212 162-00001 180-00001 188-00001 199-00001 112-RG304 5/ 171-00001			
0104 Cable group XIV	79-RG218 5/ 166-00001 193-00001			
0012 Cable group X	6-RG11 4/ 181-00001 4/ 62-RG144 4/ 65-RG165 159-00001 74-RG213 189-00001 163-00001 75-RG214 190-00001 164-00001 86-00001 6/ 127-RG393 5/ 174-00001 77-RG216 4/ 191-00001 4/	A B	.850 (21.59)	1.900 (48.26) .900 (22.86)
0102 Cable group XII	78-RG217 5/	A B	.850 (21.59)	2.094 (53.19) .900 (22.86)
0125 Cable group XI	74-RG215 5/ 7/ 189-00002 7/	A B	.850 (21.59	1.900 (48.26) .900 (22.86)
Category C - Field replaceable (MIL-C-22520 crimp tool) See footnote next to applicable cable for crimp die 3/ 8/				
0013 (Superseding -0023 13/) Cable group VIII	73-RG212 9/ 162-00001 9/ 2-RG006 4/ 9/ 180-00001 9/ 188-00001 9/ 199-00001 9/ 112-RG303 5/ 9/ 171-00001 9/			
0014 (Superseding -0018 13/) Cable group XA	74-RG213 10/ 65-RG165 5/ 6/ 10/ 159-00001 10/ 189-00001 10/ 163-00001 10/	A B	.850 (21.59)	2.031 (51.59) .900 (22.86)
0015 (Superseding -0019 13/) Cable group XB	75-RG214 10/ 190-00001 10/ 164-00001 10/ 86-00001 6/ 10/ 127-RG393 5/ 10/ 174-00001 10/			
0020 Cable group XC	6-RG11 4/ 10/ 181-00001 4/ 10/ 62-RG144 4/ 5/ 6/ 10/			
0021 Cable group XD	77-RG216 4/ 5/ 6/ 10/ 191-00001 10/			

See footnotes at end of table.

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TABLE I. Dash numbers, cross-reference, and dimensions - Continued.

Dash number 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters)	
			Minimum	Maximum
Category D - Field replaceable - Defined piece part 3/ 8/ 11/ 12/				
0501 Cable group XB	75-RG214 190-00001 164-00001 86-00001 127-RG393 5/ 174-00001	A B	.850 (21.59)	1.875 (47.62) .900 (22.86)
0502 Cable group XA	74-RG213 65-RG165 5/ 159-00001 189-00001 163-00001			
0503 Cable group VIB	60-RG142 6/ 158-00001 84-RG223 167-00001 194-00001 200-00001 128-RG400 5/ 175-00001	A B	.850 (21.59)	1.781 (45.24) .900 (22.86)
0504 Cable group VIA	28-RG058 155-00001 183-00001 197-00001 111-RG303 5/ 170-00001			

1/ For cross-reference of dash number to superseded PIN or type designation, see table IV.

2/ The latest version of each cable shall be applicable.

3/ These connectors have captivated center contacts.

4/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, Rf leakage and insertion loss are not applicable.

5/ Cable to be used when performing tests requiring cable except as in 4/ and 6/.

6/ Cable to be used for the +200°C temperature cycling tests.

7/ Armored cable.

8/ These connectors are assembled using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

9/ M22520/5-35 closure A or M22520/5-55 closure A.

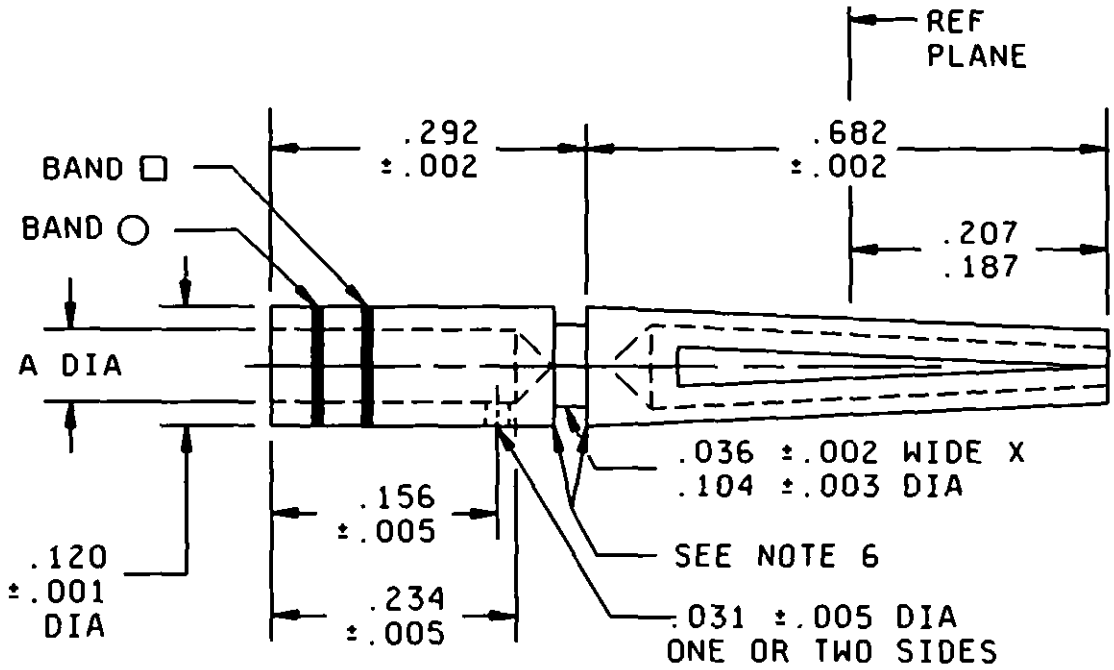
10/ M22520/5-61.

11/ Complete connector assembly shall consist of a body, center contact, ferrule and assembly instructions.

12/ Not to be used in army equipment.

13/ The superseding dash numbers are acceptable for government use until stock is purged or 3 years from the date of this specification which ever is earlier.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.043	1.09	.207	5.26	.438	11.13
.002	0.05	.098	2.49	.220	5.59	.492	12.50
.003	0.08	.104	2.64	.234	5.94	.500	12.70
.005	0.13	.120	3.05	.245	6.22	.600	15.24
.015	0.38	.156	3.96	.250	6.35	.682	17.63
.031	0.79	.187	4.75	.292	7.42		
.036	0.91	.206	5.23	.418	10.62		

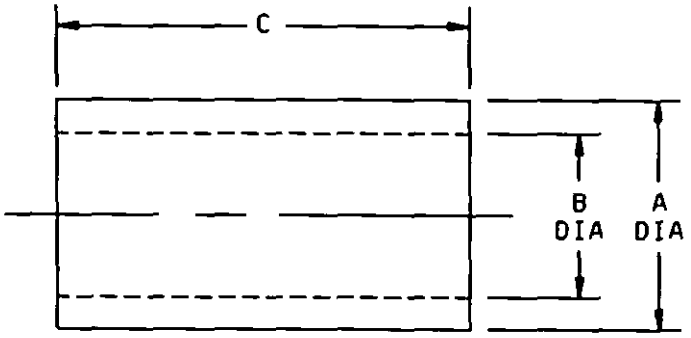
Dash no.	Contact no. 1/	Dimension A	Basic crimp tool 2/ M22520/1	Crimp die or positioner M22520/1 3/	Crimp tensile minimum (pounds)	Color band □	Color band ○
0501 0502	3-10	.098 ±.002	-01	-14	60	Red	Red
0503 0504	3-11	.043 +.001 -.002	-01	-14	20	Blue	Red

1/ Contact numbers are for identification only.
2/ Class 2 tool may be used by OEM (see MIL-C-22520).
3/ Optional tool: M22520/5-01 with M22520/5-25 closure B die.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheet which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. Maximum break of .003 inch (0.08 mm).
7. Color bands shall be positioned such that no coloring material enters the inspection hole.

FIGURE 3. Contact and ferrule dimensions for category D only.

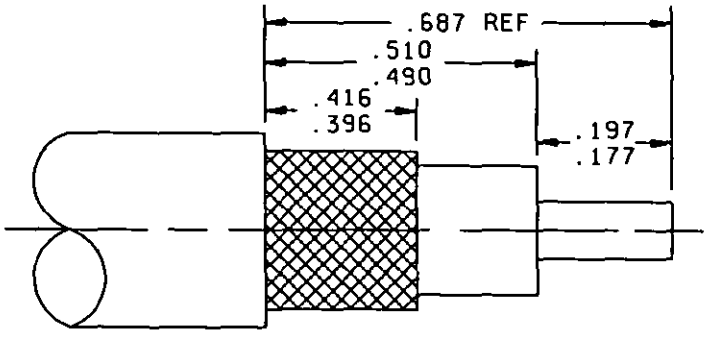
Source: https://assist.dla.mil -- Downloaded: 20260-21T16:38Z
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Dash number	Ferrule number 1/	A ±.003	B ±.003	C ±.015	Basic crimp tool 2/	Crimp die or positioner M22520/5-
0501	3-50	.492	.438	.600	M22520/5-01	25 Closure A or 61
0502	3-51	.492	.418	.600		
0503	3-52	.250	.220	.500		5, 11, 57 Closure A or 19 Closure B
0504	3-53	.245	.206	.500		

1/ Ferrule numbers are for identification only.
2/ Class 2 tools may be used by OEM (see MIL-C-22520).

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.



Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

1,000 volts rms maximum working voltage at sea level.

250 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Design and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque: 6 to 10 inch-pounds.

Mating characteristics: See figure 2 and MIL-STD-348 for dimensions.

Center contact (socket):

Oversize test pin: .074 diameter minimum (nonclosed entry contacts only).

Insertion depth: .125 minimum.

Number of insertions: 1.

Insertion force test: Steel test pin diameter .066 minimum.

Test pin finish: 16 microinches.

Insertion force: 2 pounds maximum.

Withdrawal force test: Steel test pin diameter .063 maximum.

Withdrawal force: 2 ounces minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only.) 15 pounds minimum axial force for all cables except RG400 and RG142; 6 pounds minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30 maximum.

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Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.008 +.002 F (F in GHz).

Item 16: VSWR shall be less than 1.008 +.002 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.012 +.004 F (F in GHz).

Group B inspection - VSWR shall be less than 1.024 +.007 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.08.

Connector durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.2	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,500 volts rms minimum at sea level for connectors using other than RG400 and RG142; 1,500 volts rms minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 500 volts rms minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 75 pounds minimum.

Crimp assemblies:

50 lbs minimum for cables .155-.189 OD.
60 lbs minimum for cables .190-.229 OD.
75 lbs minimum for cables .230-.249 OD.
90 lbs minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .15 dB maximum at 10 GHz.

Part or Identifying Number (PIN): M39012/03- (dash number from table I or "B" number from table III).

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TABLE II. Group qualification. 1/

Group	Submission and qualification of any of the following connectors 2/ M39012/	Qualifies the following connectors M39012/
I	02-0003 02-0101 02-0104 03-0101 03-0102	02-0003 02-0101 02-0104 03-0101 03-0102
II	02-0003 02-0006 03-0012	02-0003 02-0006 03-0012
III	02B0007 02B0008 02B0009 02B0012 02B0013 02B0015 02B0016 02B0017 03B0003 03B0004 03B0005 03B0006 03B0008 03B0009	02B0007 02B0008 02B0009 02B0012 02B0013 02B0014 02B0015 02B0016 02B0017 02B0018 03B0003 03B0004 03B0005 03B0006 03B0008 03B0009 03B0010 03B0011
IV	02-0019 02-0020 02-0021 02-0022 02-0024 02-0025 02-0027 02-0028 02-0029 03-0013 03-0014 03-0015 03-0018 03-0019	02B0007 02B0008 02B0009 02B0012 02B0013 02B0014 02B0015 02B0016 02B0017 02B0018 02-0019 02-0020 02-0021 02-0022 02-0024 02-0025 02-0026 02-0027 02-0028 02-0029 02-0030 03B0003 03B0004 03B0005 03B0008 03B0009 03B0010 03B0011 03-0013 03-0014 03-0015 03-0018 03-0019 03-0020 03-0021

See footnotes at end of table.

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TABLE II. Group qualification - Continued. 1/

Group	Submission and qualification of any of the following connectors 2/ M39012/	Qualifies the following connectors M39012/
V	02B0014 02B0018 03B0010 03B0011	02B0014 02B0018 03B0010 03B0011
VI	02-0026 02-0030 03-0020 03-0021 03-0023	02B0014 02B0018 02-0026 02-0030 03B0010 03B0011 03-0020 03-0021 03-0023
VII	03B0022	03B0022
VIII	03-0501 03-0502	03-0501 03-0502 03-0503
IX	03-0503	03-0503
X	03-0504	03-0504

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector PIN's (within the same series), the manufacturer may receive qualification approval for two or more connector PIN's by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design.

2/ For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right-hand column. The part does not necessarily have to be the part initially qualified.

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TABLE III. CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED). 1/

NOT FOR ARMY, NAVY, OR AIR FORCE USE. FOR OEM USE ONLY

PIN 2/ M39012/03B	Applicable cable 3/ M17/	Dimensions	Inches (millimeters)	
			Minimum	Maximum
0003 4/	073-RG212 5/ RG-222/U	A B	.850 (21.59)	2.031 (51.59) .900 (22.86)
0004 4/	074-RG213 5/			
0005 4/	75-RG214 5/			
0006 4/	078-RG217			
0008 4/	65-RG165 5/ 6/			
0009 4/	RG-225/U 5/ 6/			
0010 4/	62-RG144 5/ 6/ 7/ 6-RG11 7/			
0011 4/	77-RG216 5/ 7/			
0022 8/	2-RG6 5/ 7/			

1/ For maintenance replacements for category B, see table V.

2/ For cross-reference of dash number to superseded PIN or type designation, see table IV.

3/ The latest version of each cable shall be applicable.

4/ Inactive for new design.

5/ Cable to be used when performing tests requiring cable except as in 6/ and 7/.

6/ Cable to be used for the +200°C temperature cycling tests.

7/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

8/ These connectors have captivated center contacts.

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TABLE IV. Cross-reference of PIN's. 1/

Preferred PIN M39012/03	Substitute for PIN or type designation 2/
-0101	UG1590/U, M39012/03-0001
-0012	M23329/2-21, UG160F/U, M39012/03-0002
B0003	M23329/2-20, M39012/03-0003, M23329/2-25, M39012/03-0007, UG-1699/U, UG-1703/U
B0004	UG-1700/U, M39012/03-0004
B0005	M23329/2-22, UG-1701/U, M39012/03-0005
B0006	M23329/2-23, UG-1702/U, M39012/03-0006
B0008	M23329/2-26, UG-1704/U, M39012/03-0008
B0009	M23329/2-27, UG-1705/U, M39012/02-0009
B0010	M23329/2-28, UG-1706/U, M39012/03-0010
B0011	UG-1819/U, M39012/03-0017
-0012	UG-1537A/U
-0013	M39012/03-0017
-0014	
-0015	
-0018	
-0019	
-0020	
-0021	
B0022	M39012/03-0022
-0023	
-0501	
-0502	
-0503	
-0504	UG-5568/U

1/ The superseded PIN or the type designation is for cross-reference only. Where a superseded PIN or type designation is not given, none was assigned or will be assigned. The PIN M39012/03-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g., UG-18/U includes UG-18A/U, UG-18B/U, etc.

TABLE V. Maintenance replacements for category B.

Category B number 1/	Category C dash number	Category A dash number	Category D dash number
B0003	0013	0101	---
B0004	0014	0012	0502
B0005	0015	0012	0501
B0006	---	---	---
B0008	0018	0012	---
B0009	0019	0012	0501
B0010	0020	0012	---
B0011	0021	0012	---
B0022	0023	0101	---

1/ Category B connectors are for original installation only. They will not be stocked or procured by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, EA, MI
Navy - AS, MC, OS, SH
Air Force - 11, 19, 99

Preparing activity:
DLA - ES

(Project 5935-3932-03)

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